

# Su Doku

3

The Official Utterly Addictive  
Number-Placing Puzzle

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 9 |   |   |   |   |   |   | 3 |
| 2 |   | 3 |   |   | 1 |   |   |   |
| 1 | 7 |   |   | 9 | 8 |   |   |   |
|   |   | 4 |   |   |   |   |   |   |
| 9 |   |   | 6 |   | 3 |   |   | 4 |
|   |   |   |   |   |   | 5 |   |   |
|   |   |   | 4 | 2 |   |   | 3 | 6 |
|   |   |   | 9 |   |   | 4 |   | 2 |
| 6 |   |   |   |   |   |   | 1 |   |

Wayne Gould Su Doku Grand Master



**NEW YORK POST**

# Su Doku

Book 3



**NEW YORK POST**

# Su Doku

The Official Utterly Addictive  
Number-Placing Puzzle

*Compiled by Wayne Gould*

Book 3



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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   | 6 |   |   |   | 7 |   |
| 7 |   |   | 1 | 4 | 5 | 6 |   |   |
| 2 |   |   | B | C | 3 |   | 4 |   |
|   |   | 1 | 3 |   |   | 8 |   | D |
|   | 6 | A |   | 8 |   |   | 9 |   |
|   |   | 9 |   |   | 7 | 5 |   |   |
|   | 7 |   | 8 |   |   |   |   | 6 |
|   |   | 2 | 7 | 5 | 4 |   |   | 8 |
|   | 5 |   |   |   | 1 |   |   |   |

## Tips from the compiler

### Where to begin? Anywhere you can!

You could just guess where the numbers go. But if you guessed wrong – and the odds are that you would – you would get yourself in an awful mess. You would be blowing away eraser-dust for hours. It's more fun to use reason and logic to winkle out the numbers' true positions.

## **Here are some logic techniques to get you started.**

Look at the 7s in the leftmost stack of three boxes. There's a 7 in the top box and a 7 in the bottom box, but there's no 7 in the middle box. Bear in mind that the 7 in the top box is also the 7 for all of the first column. And the 7 in the bottom box is also the 7 for all of the second column. So the 7 for the middle box cannot go in columns 1 and 2. It must go in column 3. Within the middle box, column 3 already has two clues entered. In fact, there's only one free cell. That cell (marked A) is the only one that can take the 7.

## **That technique is called slicing. Now for slicing-and-dicing.**

Look at the 7s in the band across the top of the grid. The leftmost box has its 7 and so does the rightmost box, but the middle box doesn't have its 7 yet. The 7 in the righthand box accounts for all of the top row. The 7 in the lefthand box does the same for the second row, although in fact the second row of the middle box is all filled up with clues, anyway. Using our slicing technique, we know that the 7 must go in cell B or cell C.

It's time to look in the other direction. Look below the middle box, right down to the middle box at the bottom of the grid. That box has a 7, and it's in column 4. There can be only one of each number in a column, so that means the 7 for the top-middle box cannot go in cell B. It must go in cell C.

The numbers you enter become clues to help you make further progress. For example, look again at the 7 we added to cell A. You can write the 7 in, if you like, to make it more obvious that A is now 7. Using slicing-and-dicing, you should be able to add the 7 to the rightmost box in the middle band. Perhaps D stands for Destination.

If you have never solved a Su Doku puzzle before, those techniques are all you need to get started. However, as you get deeper into the book, especially as you start mixing it with the Difficult puzzles, you will need to develop other skills. The best skills – the ones you will remember, without anyone having to explain them ever again – are the ones you discover for yourself. Perhaps you may even invent a few that no one has ever described before.



# Puzzles





**Easy**

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 8 |   |   | 5 |   |   | 3 | 2 |   |
| 7 |   | 3 | 1 |   |   | 4 |   |   |
| 1 | 2 |   |   |   | 9 |   |   | 8 |
| 6 | 5 |   |   | 9 | 3 |   | 8 |   |
|   | 9 |   |   |   |   |   | 1 |   |
|   | 8 |   | 6 | 2 |   |   | 9 | 3 |
| 2 |   |   | 4 |   |   |   | 6 | 7 |
|   |   | 6 |   |   | 8 | 1 |   | 2 |
|   | 7 | 4 |   |   | 6 |   |   | 5 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 7 | 1 |   |   |   | 5 |   |   | 6 |
|   |   | 9 | 2 | 6 |   |   | 3 | 4 |
|   |   | 2 | 9 |   |   | 8 |   |   |
| 5 |   |   |   | 4 |   | 6 | 2 |   |
| 1 |   |   | 7 |   | 3 |   |   | 9 |
|   | 8 | 3 |   | 5 |   |   |   | 7 |
|   |   | 5 |   |   | 8 | 3 |   |   |
| 4 | 3 |   |   | 7 | 2 | 9 |   |   |
| 9 |   |   | 4 |   |   |   | 5 | 1 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 |   |   |   | 7 | 5 | 4 |   |
| 4 |   |   | 8 | 5 |   |   |   | 7 |
| 6 |   | 5 |   | 3 |   | 1 |   |   |
| 7 |   |   | 5 |   | 6 |   | 3 |   |
|   | 2 | 4 |   |   |   | 8 | 6 |   |
|   | 8 |   | 1 |   | 2 |   |   | 9 |
|   |   | 9 |   | 7 |   | 4 |   | 5 |
| 8 |   |   |   | 1 | 9 |   |   | 6 |
|   | 4 | 3 | 6 |   |   |   | 7 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 9 |   | 2 |   | 7 |   | 8 |   |   |
|   |   | 4 |   |   | 9 |   | 6 |   |
| 1 | 3 |   | 5 |   |   |   |   | 2 |
| 4 |   |   | 8 | 5 |   |   | 1 |   |
|   | 8 | 9 | 4 |   | 1 | 6 | 7 |   |
|   | 1 |   |   | 3 | 6 |   |   | 8 |
| 8 |   |   |   |   | 2 |   | 3 | 6 |
|   | 5 |   | 6 |   |   | 9 |   |   |
|   |   | 7 |   | 4 |   | 2 |   | 1 |





**Mild**

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   | 3 |   |   |
|   | 8 |   |   |   | 3 |   |   | 9 |
|   | 2 | 4 |   | 7 |   |   | 8 | 6 |
| 9 |   |   |   |   | 5 | 4 |   | 8 |
|   |   |   | 4 |   | 2 |   |   |   |
| 1 |   | 6 | 9 |   |   |   |   | 5 |
| 5 | 6 |   |   | 3 |   | 1 | 4 |   |
| 4 |   |   | 5 |   |   |   | 9 |   |
|   |   | 7 |   |   |   |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   | 2 | 1 |   |   |
|   | 6 | 9 |   | 4 |   |   | 7 |   |
| 4 |   |   |   | 1 |   |   | 6 |   |
| 3 |   |   | 8 |   | 5 |   |   |   |
|   | 9 | 1 |   |   |   | 7 | 3 |   |
|   |   |   | 9 |   | 1 |   |   | 5 |
|   | 7 |   |   | 5 |   |   |   | 2 |
|   | 3 |   |   | 2 |   | 4 | 1 |   |
|   |   | 2 | 4 |   |   |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 3 | 5 |   | 1 | 8 |   |   | 2 |   |
|   |   |   | 7 |   |   |   | 1 | 9 |
|   |   |   |   |   |   | 7 |   |   |
|   |   | 4 | 8 |   | 1 |   | 7 |   |
| 9 |   |   |   | 4 |   |   |   | 8 |
|   | 8 |   | 6 |   | 2 | 9 |   |   |
|   |   | 5 |   |   |   |   |   |   |
| 8 | 9 |   |   |   | 5 |   |   |   |
|   | 3 |   |   | 2 | 7 |   | 5 | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 |   |   |   |   |   | 2 |   |
|   |   | 5 | 4 |   | 1 | 9 |   |   |
| 8 |   | 6 |   | 9 |   | 7 |   | 5 |
| 7 |   |   | 3 |   | 4 |   |   | 6 |
|   | 2 |   |   |   |   |   | 7 |   |
| 4 |   |   | 7 |   | 9 |   |   | 1 |
| 9 |   | 7 |   | 5 |   | 3 |   | 8 |
|   |   | 2 | 6 |   | 8 | 1 |   |   |
|   | 8 |   |   |   |   |   | 5 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 7 |   | 8 |   | 1 |   | 9 |   | 4 |
|   |   |   |   |   |   |   |   |   |
|   | 3 | 2 |   |   |   | 6 | 7 |   |
| 4 |   |   | 1 |   | 9 |   |   | 7 |
|   |   | 6 | 2 |   | 8 | 4 |   |   |
| 2 |   |   | 5 |   | 3 |   |   | 8 |
|   | 9 | 3 |   |   |   | 2 | 1 |   |
|   |   |   |   |   |   |   |   |   |
| 6 |   | 7 |   | 9 |   | 5 |   | 3 |

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 1 |   |   |   |  |   |   | 4 | 9 |
|   |   |   | 8 |  |   |   |   |   |
| 8 | 2 |   | 7 |  |   | 6 | 1 | 3 |
|   |   |   |   |  | 9 | 5 |   | 6 |
|   | 6 |   |   |  |   |   | 8 |   |
| 5 |   | 9 | 6 |  |   |   |   |   |
| 4 | 8 | 5 |   |  | 7 |   | 9 | 2 |
|   |   |   |   |  | 2 |   |   |   |
| 7 | 9 |   |   |  |   |   |   | 4 |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 8 | 2 | 7 | 1 | 6 |   |   |
|   |   | 6 |   |   |   | 8 |   |   |
| 7 |   |   |   |   |   |   |   | 5 |
| 2 |   |   | 1 |   | 3 |   |   | 6 |
| 3 | 5 |   | 6 |   | 9 |   | 4 | 7 |
| 8 |   |   | 7 |   | 5 |   |   | 3 |
| 1 |   |   |   |   |   |   |   | 4 |
|   |   | 5 |   |   |   | 3 |   |   |
|   |   | 9 | 8 | 5 | 7 | 1 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 3 | 4 |   | 8 | 2 |   |   |
|   | 7 |   | 1 |   | 2 |   | 4 |   |
|   | 1 |   |   |   |   |   | 6 |   |
| 3 |   |   |   | 4 |   |   |   | 6 |
| 7 |   | 2 |   |   |   | 8 |   | 5 |
| 5 |   |   |   | 2 |   |   |   | 3 |
|   | 3 |   |   |   |   |   | 5 |   |
|   | 5 |   | 6 |   | 7 |   | 3 |   |
|   |   | 9 | 5 |   | 4 | 1 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 9 |   |   |   | 1 |   |   |
| 5 |   |   |   | 7 |   |   |   | 4 |
|   |   | 7 | 9 | 1 | 5 | 8 |   |   |
| 9 |   | 2 |   |   |   | 4 |   | 8 |
|   | 7 |   | 2 |   | 8 |   | 3 |   |
| 3 |   | 5 |   |   |   | 2 |   | 7 |
|   |   | 8 | 3 | 9 | 4 | 7 |   |   |
| 4 |   |   |   | 6 |   |   |   | 9 |
|   |   | 6 |   |   |   | 3 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 9 |   |   | 1 |   | 2 | 5 |   |
|   |   | 7 |   |   |   | 8 |   |   |
|   |   |   | 9 |   | 8 |   |   | 3 |
|   | 4 |   |   | 8 | 6 |   |   | 2 |
|   |   | 1 |   |   |   | 6 |   |   |
| 9 |   |   | 4 | 2 |   |   | 1 |   |
| 6 |   |   | 8 |   | 5 |   |   |   |
|   |   | 8 |   |   |   | 9 |   |   |
|   | 5 | 2 |   | 7 |   |   | 3 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 |   |   | 2 |   |   |   | 7 | 3 |
| 6 |   |   |   | 7 | 9 |   |   |   |
|   |   | 4 |   |   |   | 5 |   |   |
|   | 9 |   | 1 |   | 3 |   |   | 8 |
|   | 8 |   |   |   |   |   | 4 |   |
| 3 |   |   | 6 |   | 2 |   | 1 |   |
|   |   | 7 |   |   |   | 1 |   |   |
|   |   |   | 9 | 5 |   |   |   | 7 |
| 8 | 3 |   |   |   | 7 |   |   | 5 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 |   |   |   |   |   | 3 | 6 |
|   |   | 1 | 8 |   | 6 | 4 |   |   |
|   |   | 7 |   | 4 |   | 5 |   |   |
|   | 4 |   | 6 |   | 3 |   | 8 |   |
|   |   | 2 |   |   |   | 9 |   |   |
|   | 5 |   | 4 |   | 1 |   | 6 |   |
|   |   | 6 |   | 3 |   | 2 |   |   |
|   |   | 8 | 2 |   | 7 | 6 |   |   |
| 2 | 7 |   |   |   |   |   | 1 | 4 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 7 |   |   |   | 6 |   |   |   | 5 |
| 3 | 2 | 4 |   |   |   | 6 | 9 | 8 |
|   | 6 |   |   |   |   |   | 2 |   |
|   |   | 7 | 3 |   | 4 | 8 |   |   |
|   |   |   | 1 |   | 6 |   |   |   |
|   |   | 6 | 9 |   | 2 | 5 |   |   |
|   | 3 |   |   |   |   |   | 5 |   |
| 6 | 4 | 2 |   |   |   | 7 | 8 | 3 |
| 1 |   |   |   | 2 |   |   |   | 9 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   | 6 | 1 |   |   |   |   |
|   |   |   |   | 8 |   | 4 | 9 | 6 |
| 6 | 9 |   |   |   |   |   |   | 7 |
| 7 |   | 6 |   |   |   |   | 8 |   |
| 1 | 5 |   |   |   |   |   | 3 | 9 |
|   | 2 |   |   |   |   | 7 |   | 4 |
| 2 |   |   |   |   |   |   | 7 | 8 |
| 8 | 6 | 1 |   | 9 |   |   |   |   |
|   |   |   |   | 6 | 3 |   |   |   |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 4 |   |   |   | 3 |   |   |
|   |   |   | 8 |   |   | 9 | 4 |   |
| 2 |   |   |   | 6 |   | 5 |   | 8 |
|   |   |   |   |   | 5 |   | 9 |   |
|   | 7 | 1 |   |   |   | 2 | 8 |   |
|   | 6 |   | 9 |   |   |   |   |   |
| 1 |   | 5 |   | 2 |   |   |   | 4 |
|   | 2 | 3 |   |   | 8 |   |   |   |
|   |   | 9 |   |   |   | 8 | 6 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 6 | 4 |   |   | 1 |   |   |   |
|   |   |   |   |   | 9 | 8 |   | 6 |
|   |   |   |   | 3 | 4 |   |   |   |
| 7 |   |   |   | 4 |   |   | 8 | 3 |
|   | 5 |   |   |   |   |   | 9 |   |
| 1 | 2 |   |   | 8 |   |   |   | 5 |
|   |   |   | 4 | 7 |   |   |   |   |
| 8 |   | 2 | 5 |   |   |   |   |   |
|   |   |   | 2 |   |   | 5 | 3 | 7 |

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
|   |   |   | 3 |  |   |   |   | 8 |
| 5 | 3 |   | 7 |  |   |   |   | 4 |
|   |   | 4 |   |  | 8 | 2 |   | 9 |
|   | 5 | 1 |   |  |   |   |   |   |
|   | 9 |   |   |  |   |   | 8 |   |
|   |   |   |   |  |   | 3 | 5 |   |
| 4 |   | 7 | 8 |  |   | 6 |   |   |
| 6 |   |   |   |  | 2 |   | 1 | 7 |
| 3 |   |   |   |  | 4 |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 3 |   |   |   | 5 | 7 |   |
| 9 |   |   | 5 |   | 6 |   |   | 1 |
| 6 |   |   |   |   |   |   |   | 3 |
|   | 7 |   |   | 8 |   |   | 5 |   |
| 1 |   | 9 |   | 6 |   | 8 |   | 7 |
|   | 6 |   |   | 4 |   |   | 3 |   |
| 8 |   |   |   |   |   |   |   | 2 |
| 2 |   |   | 8 |   | 7 |   |   | 5 |
|   | 3 | 6 |   |   |   | 9 | 1 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 2 | 5 | 8 |   |   |   |   |   | 9 |
|   |   |   | 9 | 1 |   |   |   | 2 |
|   |   |   |   |   | 8 |   |   | 7 |
|   |   | 4 |   | 8 |   |   | 9 |   |
|   | 7 |   | 1 | 6 | 3 |   | 4 |   |
|   | 2 |   |   | 9 |   | 5 |   |   |
| 3 |   |   | 4 |   |   |   |   |   |
| 1 |   |   |   | 7 | 5 |   |   |   |
| 7 |   |   |   |   |   | 1 | 2 | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 |   |   |   | 3 | 6 |   | 8 |
|   |   |   | 1 |   |   |   |   | 5 |
|   |   | 3 |   | 2 |   | 9 |   | 4 |
| 3 |   | 2 | 5 |   |   |   |   |   |
| 8 |   |   |   |   |   |   |   | 2 |
|   |   |   |   |   | 4 | 1 |   | 3 |
| 1 |   | 7 |   | 5 |   | 8 |   |   |
| 2 |   |   |   |   | 9 |   |   |   |
| 9 |   | 5 | 3 |   |   |   | 7 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 4 | 6 | 9 |   |   |   |   |   |   |
| 5 | 3 |   |   |   | 7 | 8 |   |   |
|   |   |   | 2 |   |   |   | 6 |   |
|   |   |   | 7 | 1 | 3 |   |   | 8 |
| 8 |   |   |   |   |   |   |   | 3 |
| 6 |   |   | 5 | 4 | 8 |   |   |   |
|   | 2 |   |   |   | 9 |   |   |   |
|   |   | 4 | 6 |   |   |   | 5 | 7 |
|   |   |   |   |   |   | 2 | 3 | 9 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 3 | 6 | 1 |   |   |   |   |   |
|   |   |   |   | 7 |   |   |   | 3 |
|   |   | 7 | 5 | 4 |   | 2 |   | 1 |
|   |   |   |   |   |   | 7 |   | 5 |
|   | 7 | 3 |   |   |   | 8 | 1 |   |
| 2 |   | 8 |   |   |   |   |   |   |
| 7 |   | 2 |   | 1 | 9 | 4 |   |   |
| 6 |   |   |   | 2 |   |   |   |   |
|   |   |   |   |   | 6 | 1 | 5 |   |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |
| 1 |   |   | 3 |   | 4 |   |   | 9 |
|   | 7 | 3 | 5 | 9 | 2 | 6 | 8 |   |
|   |   | 9 |   | 4 |   | 1 |   |   |
| 6 |   | 8 |   |   |   | 7 |   | 5 |
|   |   | 4 |   | 7 |   | 8 |   |   |
|   | 6 | 2 | 1 | 8 | 9 | 4 | 3 |   |
| 9 |   |   | 7 |   | 3 |   |   | 8 |
|   |   |   |   |   |   |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 7 |   |   |   |   |   | 5 |   |
|   |   | 8 |   |   |   | 9 |   |   |
| 4 |   |   | 2 |   | 1 |   |   | 8 |
| 2 | 8 |   | 4 | 5 | 3 |   | 9 | 7 |
|   |   |   |   |   |   |   |   |   |
| 1 | 3 |   | 6 | 8 | 7 |   | 4 | 2 |
| 6 |   |   | 5 |   | 9 |   |   | 3 |
|   |   | 1 |   |   |   | 4 |   |   |
|   | 5 |   |   |   |   |   | 6 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   | 3 |   |   | 9 |   |   |
|   |   |   |   |   | 6 | 1 |   |   |
|   | 6 | 7 | 9 | 5 |   |   |   | 8 |
| 2 |   |   |   | 9 |   |   | 5 | 3 |
| 3 |   |   |   |   |   |   |   | 9 |
| 8 | 1 |   |   | 6 |   |   |   | 7 |
| 1 |   |   |   | 2 | 8 | 7 | 3 |   |
|   |   | 2 | 6 |   |   |   |   |   |
|   |   | 3 |   |   | 5 |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   | 5 |   |   |   | 9 |
|   | 5 |   |   |   | 7 | 2 |   |   |
|   |   |   |   |   | 8 | 6 | 5 |   |
| 9 |   | 4 | 7 |   |   |   |   |   |
|   | 6 | 8 | 4 |   | 1 | 9 | 7 |   |
|   |   |   |   |   | 6 | 8 |   | 5 |
|   | 7 | 1 | 8 |   |   |   |   |   |
|   |   | 6 | 5 |   |   |   | 9 |   |
| 4 |   |   |   | 3 |   |   |   |   |



**Difficult**

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| 7 |   |  |   |   |   |  |   | 9 |
|   |   |  | 1 | 8 | 3 |  |   |   |
| 5 | 3 |  |   |   |   |  | 6 | 4 |
|   | 6 |  | 9 |   | 8 |  | 3 |   |
| 1 |   |  |   |   |   |  |   | 5 |
|   | 5 |  | 2 |   | 7 |  | 4 |   |
| 3 | 7 |  |   |   |   |  | 1 | 8 |
|   |   |  | 8 | 2 | 6 |  |   |   |
| 6 |   |  |   |   |   |  |   | 2 |

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|   | 2 |   |   |   |   |   |   |   |
|   |   | 9 | 6 | 3 |   | 5 | 8 |   |
|   |   |   |   |   | 7 |   |   |   |
| 9 |   |   |   | 4 |   |   |   | 3 |
| 1 |   |   | 8 |   | 5 |   |   | 9 |
| 7 |   |   |   | 2 |   |   |   | 6 |
|   |   |   | 4 |   |   |   |   |   |
|   | 3 | 5 |   | 9 | 2 | 4 |   |   |
|   |   |   |   |   |   |   | 7 |   |



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|   | 3 |   | 7 |   |   |   |   | 1 |
|   |   |   |   |   |   | 7 | 2 |   |
|   | 4 |   |   | 6 | 1 |   | 9 |   |
|   |   | 5 |   |   |   | 6 |   | 3 |
|   |   |   | 6 |   | 4 |   |   |   |
| 6 |   | 1 |   |   |   | 2 |   |   |
|   | 6 |   | 1 | 2 |   |   | 7 |   |
|   | 9 | 8 |   |   |   |   |   |   |
| 1 |   |   |   |   | 9 |   | 8 |   |

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|   |   |   | 2 | 8 | 5 |   | 3 |   |
|   | 4 |   |   |   |   |   | 9 |   |
| 8 |   |   |   | 7 |   |   |   |   |
| 2 |   |   |   |   | 6 |   |   |   |
| 5 |   | 4 |   |   |   | 1 |   | 9 |
|   |   |   | 1 |   |   |   |   | 4 |
|   |   |   |   | 4 |   |   |   | 2 |
|   | 7 |   |   |   |   |   | 4 |   |
|   | 1 |   | 5 | 2 | 8 |   |   |   |

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|   |   |   | 5 |   |   | 6 |   |   |
|   |   |   | 9 |   |   |   | 2 |   |
| 1 | 5 | 2 |   |   |   | 9 |   |   |
|   |   | 9 |   | 2 |   |   |   | 8 |
|   |   |   | 4 |   | 8 |   |   |   |
| 6 |   |   |   | 7 |   | 3 |   |   |
|   |   | 7 |   |   |   | 1 | 8 | 6 |
|   | 2 |   |   |   | 7 |   |   |   |
|   |   | 8 |   |   | 5 |   |   |   |

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|   |   |   |   |   |   |   |   |   |
|   |   | 8 | 3 |   | 4 | 7 |   |   |
| 3 | 7 |   |   |   |   |   | 9 | 8 |
| 9 | 6 |   |   | 2 |   |   | 7 | 5 |
|   |   |   |   | 1 |   |   |   |   |
| 5 | 4 |   |   | 9 |   |   | 3 | 6 |
| 4 | 2 |   |   |   |   |   | 8 | 7 |
|   |   | 3 | 8 |   | 5 | 6 |   |   |
|   |   |   |   |   |   |   |   |   |

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|   | 8 |   |   | 4 |   | 6 |   |   |
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| 9 | 6 |   |   |   |   |   |   |   |
|   |   |   |   | 6 |   |   | 5 |   |
| 3 |   |   | 1 |   | 2 |   |   | 7 |
|   | 4 |   |   | 3 |   |   |   |   |
|   |   |   |   |   |   |   | 9 | 1 |
| 5 |   | 2 |   |   | 9 |   |   |   |
|   |   | 8 |   | 2 |   |   | 4 |   |

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|   |   | 9 | 5 |   | 2 |   | 8 |   |
|   |   | 1 |   | 8 |   |   |   | 9 |
|   | 4 |   |   |   |   | 1 |   |   |
| 1 |   |   |   |   | 6 | 4 |   |   |
|   | 9 |   |   |   |   |   | 1 |   |
|   |   | 4 | 1 |   |   |   |   | 2 |
|   |   | 5 |   |   |   |   | 2 |   |
| 7 |   |   |   | 5 |   | 6 |   |   |
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| 9 | 8 |   |   |   |   |   |   |   |
|   | 7 | 1 |   |   |   |   |   | 2 |
|   |   | 5 |   | 4 |   |   |   |   |
|   | 3 | 4 |   |   | 7 |   |   |   |
|   | 1 |   | 4 |   | 6 |   | 8 |   |
|   |   |   | 1 |   |   | 6 | 5 |   |
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|   |   |   |   | 2 |   |   | 8 |   |
|   |   | 4 |   |   | 9 | 5 | 2 |   |
|   |   | 7 | 4 | 6 |   |   | 5 | 9 |
|   |   | 3 |   |   |   | 1 |   |   |
| 5 | 6 |   |   | 9 | 8 | 4 |   |   |
|   | 3 | 9 | 1 |   |   | 7 |   |   |
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|   | 4 | 7 | 8 |   |   |   |   |   |
| 8 | 1 |   |   |   |   |   | 6 | 7 |
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|   |   |   |   | 4 | 3 |   |   |   |
|   | 5 | 4 |   | 7 |   | 1 | 8 |   |
|   |   |   | 5 | 9 |   |   |   |   |
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| 5 | 3 |   |   |   |   |   | 9 | 1 |
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|   | 4 |   |   | 5 |   |   | 3 |   |
| 9 | 5 |   |   |   |   |   | 1 | 2 |
|   |   | 2 |   |   |   | 6 |   |   |
| 8 |   |   | 5 |   | 6 |   |   | 9 |
|   |   |   |   |   |   |   |   |   |
| 1 |   |   | 7 |   | 4 |   |   | 5 |
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|   |   |   |   |   | 9 |   |   |   |
| 1 |   | 6 |   |   | 8 |   |   |   |
|   |   |   |   | 7 | 4 | 2 |   | 9 |
| 5 |   |   |   |   |   |   | 4 |   |
|   |   | 7 | 2 |   | 1 | 5 |   |   |
|   | 8 |   |   |   |   |   |   | 1 |
| 8 |   | 4 | 5 | 1 |   |   |   |   |
|   |   |   | 4 |   |   | 6 |   | 8 |
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| 5 |   | 1 |   |   |   | 9 |   | 6 |
| 7 |   |   |   | 6 |   |   |   | 3 |
|   |   |   | 7 |   | 2 |   |   |   |
|   |   | 3 |   | 1 |   | 5 |   |   |
|   | 5 |   |   | 8 |   |   | 4 |   |
|   |   | 8 |   | 2 |   | 7 |   |   |
|   |   |   | 8 |   | 6 |   |   |   |
| 8 |   |   |   | 7 |   |   |   | 9 |
| 6 |   | 9 |   |   |   | 3 |   | 5 |

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|   | 5 |   | 4 |   |   | 2 |   | 9 |
|   |   |   |   |   | 5 |   |   | 4 |
|   | 7 |   |   | 6 |   |   |   |   |
| 9 |   |   |   |   |   |   | 5 | 8 |
|   |   | 4 |   | 7 |   | 3 |   |   |
| 8 | 2 |   |   |   |   |   |   | 7 |
|   |   |   |   | 9 |   |   | 6 |   |
| 6 |   |   | 7 |   |   |   |   |   |
| 1 |   | 2 |   |   | 6 |   | 3 |   |

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|   |   | 1 |   |   |   | 3 |   |   |
|   |   | 2 |   |   |   | 7 |   |   |
|   |   |   | 8 | 2 | 3 |   |   |   |
| 2 |   |   |   | 9 |   |   |   | 6 |
|   | 1 |   | 4 |   | 5 |   | 7 |   |
| 3 |   |   |   | 7 |   |   |   | 4 |
|   |   |   | 7 | 6 | 2 |   |   |   |
|   |   | 7 |   |   |   | 5 |   |   |
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| 9 | 3 |   |   | 6 |   |   |   | 5 |
|   | 1 |   |   |   |   |   | 3 | 8 |
|   |   |   | 8 |   |   |   |   |   |
|   |   |   | 5 |   | 2 | 8 |   |   |
| 7 |   |   |   |   |   |   |   | 9 |
|   |   | 2 | 1 |   | 4 |   |   |   |
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| 1 | 4 |   |   |   |   |   | 2 |   |
| 5 |   |   |   | 2 |   |   | 8 | 7 |

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|   |   | 7 |   |   | 4 |   |   | 8 |
|   |   | 8 |   |   |   |   |   | 3 |
|   |   |   |   | 6 | 1 | 2 |   |   |
|   |   |   |   | 4 |   |   | 7 |   |
| 2 |   | 4 |   |   |   | 3 |   | 9 |
|   | 9 |   |   | 2 |   |   |   |   |
|   |   | 6 | 1 | 9 |   |   |   |   |
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|   | 3 |   |   |   |   |   |   | 5 |
|   | 8 | 9 |   | 3 |   | 1 |   |   |
| 2 |   |   |   |   |   |   |   |   |
| 9 |   |   |   |   | 7 | 5 |   | 6 |
|   |   |   | 6 |   | 5 |   |   |   |
| 5 |   | 2 | 3 |   |   |   |   | 4 |
|   |   |   |   |   |   |   |   | 9 |
|   |   | 7 |   | 1 |   | 2 | 3 |   |
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|   | 9 |   |   |   |   |   |   | 3 |
| 2 |   | 3 |   |   | 1 |   |   |   |
| 1 | 7 |   |   | 9 | 8 |   |   |   |
|   |   | 4 |   |   |   |   |   |   |
| 9 |   |   | 6 |   | 3 |   |   | 4 |
|   |   |   |   |   |   | 5 |   |   |
|   |   |   | 4 | 2 |   |   | 3 | 6 |
|   |   |   | 9 |   |   | 4 |   | 2 |
| 6 |   |   |   |   |   |   | 1 |   |

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|   |   |   |   | 1 |   |   |   | 4 |
|   |   |   | 9 | 8 |   | 7 | 2 |   |
|   |   |   |   |   | 4 |   | 5 |   |
|   |   | 7 |   |   |   |   |   | 6 |
| 8 | 6 |   |   | 5 |   |   | 7 | 1 |
| 2 |   |   |   |   |   | 8 |   |   |
|   | 7 |   | 3 |   |   |   |   |   |
|   | 3 | 9 |   | 4 | 8 |   |   |   |
| 5 |   |   |   | 2 |   |   |   |   |

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| 6 | 8 |   |   |   |   |   |   |   |
|   | 1 |   | 7 | 4 | 2 | 6 |   |   |
|   |   | 3 |   |   |   |   |   |   |
|   |   |   | 2 |   |   |   |   | 7 |
| 8 | 7 |   | 4 |   | 5 |   | 3 | 2 |
| 4 |   |   |   |   | 8 |   |   |   |
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|   |   | 1 | 8 | 9 | 6 |   | 7 |   |
|   |   |   |   |   |   |   | 1 | 4 |

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|   | 8 |   | 9 |   | 4 |   | 1 |   |
|   |   | 3 |   | 7 |   | 2 |   |   |
|   |   |   | 6 |   | 8 |   |   |   |
| 7 |   |   |   | 1 |   |   |   | 2 |
|   |   | 2 |   |   |   | 7 |   |   |
| 3 |   |   |   | 5 |   |   |   | 1 |
|   |   |   | 8 |   | 7 |   |   |   |
|   |   | 1 |   | 4 |   | 5 |   |   |
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|   | 9 | 4 |   |   |   |   |   |   |
| 1 |   |   | 7 | 9 |   |   |   |   |
|   |   | 3 |   | 8 | 6 |   |   | 1 |
| 9 | 2 |   |   |   |   |   | 3 |   |
|   |   |   |   | 7 |   |   |   |   |
|   | 8 |   |   |   |   |   | 2 | 5 |
| 4 |   |   | 5 | 2 |   | 9 |   |   |
|   |   |   |   | 6 | 1 |   |   | 3 |
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|   | 5 |   |   | 1 |   |   |   |   |
|   | 2 |   |   |   | 3 |   |   | 7 |
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| 3 |   |   |   |   |   |   |   | 1 |
|   |   | 7 |   | 9 |   |   |   | 8 |
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|   | 3 |   |   |   | 6 | 9 |   |   |

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|   |   | 5 | 2 |   |   |   | 4 |   |
| 9 |   |   |   |   |   |   |   | 5 |
|   | 2 |   |   |   | 8 | 6 |   |   |
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|   | 5 | 9 |   |   |   | 7 |   |   |
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|   |   |   |   | 3 | 9 |   | 7 | 6 |
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| 1 | 3 |   | 8 | 5 |   |   |   |   |
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|   | 8 |   |   | 3 |   |   | 9 |   |
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| 6 |   | 2 |   |   |   | 1 |   | 8 |
|   | 7 |   |   | 8 |   |   | 2 |   |
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| 3 | 2 |   |   |   |   |   | 4 | 9 |
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|   | 7 | 1 | 9 |   | 8 | 3 | 5 |   |
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| 4 |   |   |   |   |   |   |   | 1 |
|   | 1 |   | 6 |   | 5 |   | 9 |   |
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|   | 9 | 8 | 2 |   | 1 | 7 | 4 |   |
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|   | 5 |   | 6 |   | 1 |   | 8 |   |
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|   |   | 6 |   | 7 |   | 8 |   |   |
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|   | 7 | 8 | 6 |   | 1 | 5 | 2 |   |
|   | 3 |   | 4 |   | 5 |   | 1 |   |
| 4 |   |   |   |   |   |   |   | 2 |
|   | 9 |   | 3 |   | 2 |   | 6 |   |
|   | 5 | 2 | 1 |   | 3 | 6 | 9 |   |
|   |   |   |   |   |   |   |   |   |
|   |   | 3 |   | 2 |   | 1 |   |   |

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| 1 |   | 6 |   | 3 |   |   |   |   |
| 3 |   | 7 |   |   | 9 |   |   |   |
|   |   | 8 |   | 6 |   | 2 |   |   |
|   |   |   | 8 |   |   | 4 |   | 7 |
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|   |   | 5 |   |   | 3 |   | 7 |   |
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|   |   | 8 | 3 |  | 1 | 9 |   |   |
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| 7 |   | 4 |   |  |   | 8 |   | 9 |
| 2 |   |   | 4 |  | 6 |   |   | 5 |
| 6 |   | 1 |   |  |   | 2 |   | 3 |
|   | 1 |   |   |  |   |   | 2 |   |
|   |   |   | 1 |  | 8 |   |   |   |
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|   |   |   |   |   |   | 4 |   |   |
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|   |   | 9 |   |   | 3 |   |   | 5 |
|   |   |   | 9 |   | 8 |   |   |   |
| 8 |   |   | 5 |   |   | 9 |   |   |
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|   |   | 1 |   |   |   | 4 |   |   |
| 4 | 6 |   |   | 1 |   |   | 9 | 5 |
| 8 |   |   | 1 |   | 9 |   |   | 7 |
|   |   | 7 |   |   |   | 5 |   |   |
| 2 |   |   | 3 |   | 5 |   |   | 4 |
| 1 | 2 |   |   | 9 |   |   | 4 | 6 |
|   |   | 8 |   |   |   | 9 |   |   |
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|   |   |   | 6 | 5 |   |   |   | 7 |
|   | 8 |   |   | 4 |   | 6 |   |   |
|   |   | 2 | 9 |   | 7 | 1 |   |   |
|   |   | 7 |   | 2 |   |   | 9 |   |
| 4 |   |   |   | 7 | 6 |   |   |   |
|   |   |   | 4 |   |   |   |   |   |
| 9 |   |   |   |   |   | 3 |   | 5 |

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|   |   |   |   |   | 8 | 4 |   | 2 |
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|   |   | 5 |   |   |   | 6 |   |   |
|   | 7 | 3 |   |   |   |   |   |   |
|   |   |   |   | 6 |   |   | 2 | 1 |
| 4 |   | 7 | 2 |   |   |   |   |   |
| 3 |   |   | 7 |   |   |   |   | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 6 |   | 3 |   | 9 |   | 1 |   |
|   |   | 3 |   | 7 |   | 8 |   |   |
|   |   |   | 4 |   | 5 |   |   |   |
| 7 |   | 2 |   |   |   | 5 |   | 9 |
|   | 5 |   |   |   |   |   | 6 |   |
| 6 |   | 1 |   |   |   | 2 |   | 7 |
|   |   |   | 2 |   | 6 |   |   |   |
|   |   | 8 |   | 1 |   | 6 |   |   |
|   | 4 |   | 8 |   | 3 |   | 2 |   |

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
|   |   | 7 |   |  | 4 |   |   |   |
|   |   |   |   |  | 5 |   |   |   |
| 5 | 6 |   |   |  |   | 1 |   | 9 |
| 2 | 8 |   | 7 |  |   |   |   |   |
| 9 |   | 4 |   |  |   | 3 |   | 6 |
|   |   |   |   |  | 1 |   | 9 | 8 |
| 7 |   | 2 |   |  |   |   | 3 | 1 |
|   |   |   | 6 |  |   |   |   |   |
|   |   |   | 4 |  |   | 7 |   |   |

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|---|---|---|---|---|---|---|---|---|
|   |   | 7 |   |   | 2 |   |   |   |
| 5 | 8 |   |   |   | 7 |   |   |   |
|   | 4 |   | 8 |   |   | 6 |   | 7 |
| 6 | 9 | 2 |   |   |   |   |   |   |
|   |   |   |   | 3 |   |   |   |   |
|   |   |   |   |   |   | 2 | 9 | 1 |
| 7 |   | 3 |   |   | 4 |   | 5 |   |
|   |   |   | 7 |   |   |   | 8 | 9 |
|   |   |   | 6 |   |   | 4 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   | 5 | 1 |   | 7 |
| 4 | 9 |   |   |   |   |   |   |   |
|   |   | 8 | 2 | 9 |   |   |   |   |
|   | 5 |   |   | 6 |   |   |   |   |
| 1 |   |   | 7 |   | 8 |   |   | 3 |
|   |   |   |   | 3 |   |   | 2 |   |
|   |   |   |   | 1 | 6 | 9 |   |   |
|   |   |   |   |   |   |   | 6 | 8 |
| 8 |   | 5 | 9 |   |   |   |   |   |

|   |   |  |   |   |   |  |   |   |
|---|---|--|---|---|---|--|---|---|
| 7 |   |  | 3 | 6 |   |  |   |   |
| 5 |   |  |   |   |   |  |   |   |
|   | 9 |  | 2 |   | 4 |  |   | 6 |
|   |   |  | 8 |   |   |  | 7 | 9 |
|   | 6 |  |   |   |   |  | 1 |   |
| 4 | 7 |  |   |   | 5 |  |   |   |
| 8 |   |  | 1 |   | 3 |  | 4 |   |
|   |   |  |   |   |   |  |   | 2 |
|   |   |  |   | 9 | 7 |  |   | 3 |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 |   | 7 | 2 |   |   |   | 8 | 3 |
|   |   |   |   |   |   |   |   |   |
|   |   | 8 |   |   | 9 |   | 6 |   |
|   |   |   | 7 | 9 |   |   | 3 |   |
|   |   | 1 |   |   |   | 2 |   |   |
|   | 8 |   |   | 5 | 2 |   |   |   |
|   | 2 |   | 5 |   |   | 4 |   |   |
|   |   |   |   |   |   |   |   |   |
| 6 | 4 |   |   |   | 1 | 7 |   | 2 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 1 |   |   | 2 |   | 4 |   |
| 3 |   |   | 5 |   |   | 6 |   |   |
|   | 9 |   |   |   |   |   |   | 7 |
| 9 |   |   |   | 1 |   |   | 8 |   |
|   |   |   | 2 | 4 | 8 |   |   |   |
|   | 2 |   |   | 6 |   |   |   | 3 |
| 4 |   |   |   |   |   |   | 3 |   |
|   |   | 3 |   |   | 1 |   |   | 8 |
|   | 5 |   | 7 |   |   | 9 |   |   |

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|---|---|---|---|---|---|---|---|---|
| 4 |   | 5 |   |   |   | 1 |   | 9 |
|   | 1 |   |   |   | 9 |   | 3 |   |
|   |   |   | 6 |   |   |   |   |   |
| 6 | 7 |   |   | 5 |   |   |   |   |
|   | 8 |   | 2 |   | 6 |   | 9 |   |
|   |   |   |   | 9 |   |   | 1 | 5 |
|   |   |   |   |   | 8 |   |   |   |
|   | 5 |   | 9 |   |   |   | 2 |   |
| 1 |   | 3 |   |   |   | 4 |   | 6 |

# Fiendish

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 8 |   | 1 |   |   |   | 5 |   |   |
| 2 |   |   |   |   | 9 |   | 4 |   |
| 5 | 3 |   |   |   |   |   |   |   |
|   |   |   | 2 | 9 |   |   |   |   |
|   |   | 4 |   |   |   | 1 |   |   |
|   |   |   |   | 4 | 1 |   |   |   |
|   |   |   |   |   |   |   | 9 | 2 |
|   | 5 |   | 7 |   |   |   |   | 8 |
|   |   | 8 |   |   |   | 4 |   | 3 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 1 |   |   |   |   |   |   |
|   |   | 3 | 2 |   |   |   |   | 4 |
| 9 |   |   |   | 3 |   |   | 2 |   |
|   |   | 7 |   |   | 5 |   | 1 | 3 |
|   |   | 9 |   |   |   | 8 |   |   |
| 3 | 5 |   | 6 |   |   | 9 |   |   |
|   | 9 |   |   | 4 |   |   |   | 6 |
| 4 |   |   |   |   | 8 | 5 |   |   |
|   |   |   |   |   |   | 3 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 |   |   | 6 |   |   |   |   |   |
| 8 |   |   |   |   | 3 | 7 | 6 |   |
|   |   | 2 |   |   | 1 |   |   |   |
|   |   |   |   | 3 |   | 5 | 7 | 1 |
|   |   |   |   |   |   |   |   |   |
| 2 | 7 | 5 |   | 6 |   |   |   |   |
|   |   |   | 2 |   |   | 3 |   |   |
|   | 3 | 1 | 4 |   |   |   |   | 2 |
|   |   |   |   |   | 9 |   |   | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 9 |   |   | 2 |   |   |   |   | 6 |
| 1 |   |   | 5 |   |   | 7 |   |   |
|   | 8 | 3 |   |   |   |   |   |   |
|   |   |   |   | 3 |   | 8 |   |   |
| 3 |   |   | 7 |   | 1 |   |   | 9 |
|   |   | 9 |   | 5 |   |   |   |   |
|   |   |   |   |   |   | 1 | 9 |   |
|   |   | 1 |   |   | 2 |   |   | 8 |
| 2 |   |   |   |   | 8 |   |   | 5 |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 4 |   |   |   | 7 |   | 1 |
|   |   |   |   | 3 |   |   |   |   |
|   | 7 |   |   | 8 | 5 |   |   |   |
|   |   |   | 5 |   |   |   |   | 3 |
|   | 8 | 5 |   | 2 |   | 6 | 7 |   |
| 6 |   |   |   |   | 8 |   |   |   |
|   |   |   | 2 | 4 |   |   | 3 |   |
|   |   |   |   | 9 |   |   |   |   |
| 4 |   | 9 |   |   |   | 8 |   |   |

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|---|---|---|---|---|---|---|---|---|
| 6 | 5 |   |   | 9 |   |   | 2 |   |
|   |   |   |   |   | 7 |   |   |   |
| 2 |   | 9 |   |   |   |   |   |   |
|   |   | 7 |   |   | 4 |   |   |   |
|   |   | 5 | 7 |   | 6 | 8 |   |   |
|   |   |   | 2 |   |   | 1 |   |   |
|   |   |   |   |   |   | 3 |   | 7 |
|   |   |   | 6 |   |   |   |   |   |
|   | 3 |   |   | 1 |   |   | 9 | 8 |

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|---|---|---|---|---|---|---|---|---|
|   |   | 1 |   |   |   |   |   |   |
| 5 |   |   |   | 4 |   |   |   |   |
|   | 2 |   | 3 | 1 | 8 |   |   |   |
|   | 8 | 6 | 2 |   |   |   |   | 4 |
|   | 7 |   |   |   |   |   | 5 |   |
| 4 |   |   |   |   | 3 | 8 | 6 |   |
|   |   |   | 4 | 8 | 6 |   | 9 |   |
|   |   |   |   | 9 |   |   |   | 2 |
|   |   |   |   |   |   | 7 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 |   |   |   | 5 |   |   | 4 | 3 |
|   |   | 8 | 2 |   |   |   | 1 |   |
|   | 5 |   |   |   |   | 6 |   |   |
|   |   |   |   |   | 1 |   | 9 |   |
| 4 |   |   |   |   |   |   |   | 7 |
|   | 2 |   | 3 |   |   |   |   |   |
|   |   | 4 |   |   |   |   | 8 |   |
|   | 9 |   |   |   | 7 | 5 |   |   |
| 8 | 7 |   |   | 2 |   |   |   | 4 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   | 6 |   |   | 9 |   |   |
|   | 4 |   | 5 | 1 |   |   |   | 6 |
|   | 9 |   |   | 8 | 2 |   |   |   |
| 5 |   | 2 |   |   |   |   |   |   |
| 3 |   |   |   |   |   |   |   | 1 |
|   |   |   |   |   |   | 8 |   | 2 |
|   |   |   | 1 | 9 |   |   | 4 |   |
| 6 |   |   |   | 2 | 3 |   | 5 |   |
|   |   | 9 |   |   | 6 |   |   |   |

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|---|--|---|---|--|---|---|--|---|
|   |  |   |   |  |   |   |  |   |
|   |  | 4 | 6 |  | 5 | 8 |  |   |
| 8 |  | 2 |   |  |   | 5 |  | 1 |
| 3 |  |   | 2 |  | 8 |   |  | 7 |
|   |  |   | 4 |  | 6 |   |  |   |
| 2 |  |   | 5 |  | 1 |   |  | 6 |
| 7 |  | 6 |   |  |   | 4 |  | 9 |
|   |  | 8 | 3 |  | 9 | 6 |  |   |
|   |  |   |   |  |   |   |  |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   | 4 | 7 | 5 |   |   |   |
| 4 |   |   | 2 |   | 8 |   |   | 9 |
|   | 7 |   |   |   |   |   | 2 |   |
| 3 |   |   |   |   |   |   |   | 1 |
|   | 1 | 5 |   |   |   | 3 | 4 |   |
| 2 |   |   |   |   |   |   |   | 8 |
|   | 6 |   |   |   |   |   | 8 |   |
| 1 |   |   | 3 |   | 7 |   |   | 4 |
|   |   |   | 8 | 1 | 2 |   |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 8 |   |   |   |   | 2 |   | 4 |   |
|   |   |   |   | 8 |   |   |   | 3 |
|   |   |   |   |   |   | 1 | 6 |   |
|   |   |   |   |   |   |   | 3 | 1 |
|   |   | 5 | 9 | 6 | 7 | 2 |   |   |
| 2 | 4 |   |   |   |   |   |   |   |
|   | 3 | 1 |   |   |   |   |   |   |
| 6 |   |   |   | 9 |   |   |   |   |
|   | 5 |   | 1 |   |   |   |   | 7 |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 9 |   | 4 |   |   |   |   | 2 |   |
|   |   | 5 | 9 |   |   |   |   |   |
|   | 2 |   |   |   | 8 |   |   | 6 |
|   |   | 9 |   |   |   |   | 5 | 8 |
|   |   |   |   | 8 |   |   |   |   |
| 6 | 4 |   |   |   |   | 1 |   |   |
| 1 |   |   | 2 |   |   |   | 9 |   |
|   |   |   |   |   | 1 | 7 |   |   |
|   | 7 |   |   |   |   | 6 |   | 4 |

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|---|---|---|---|---|---|---|---|---|
|   |   | 3 |   |   |   | 1 |   |   |
| 5 | 8 |   |   |   |   |   | 4 | 2 |
|   |   |   | 9 |   | 7 |   |   |   |
|   | 4 |   |   | 3 |   |   | 1 |   |
|   |   |   | 2 | 8 | 4 |   |   |   |
|   | 2 |   |   | 7 |   |   | 3 |   |
|   |   |   | 6 |   | 8 |   |   |   |
| 7 | 1 |   |   |   |   |   | 6 | 3 |
|   |   | 8 |   |   |   | 5 |   |   |

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|---|---|---|---|---|---|---|---|---|
|   | 8 | 3 | 1 |   |   |   | 5 |   |
|   |   |   |   |   |   | 2 | 6 |   |
| 5 |   |   | 9 | 4 |   |   |   |   |
| 9 | 7 |   |   |   |   |   |   |   |
|   |   |   | 2 |   | 3 |   |   |   |
|   |   |   |   |   |   |   | 7 | 8 |
|   |   |   |   | 8 | 1 |   |   | 9 |
|   | 4 | 6 |   |   |   |   |   |   |
|   | 3 |   |   |   | 4 | 6 | 1 |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   | 9 |   | 3 |   |
|   |   | 8 |   | 1 |   |   |   | 7 |
|   | 6 |   |   | 3 |   |   |   | 2 |
| 7 | 9 |   | 5 |   |   |   |   |   |
|   |   | 4 |   |   |   | 1 |   |   |
|   |   |   |   |   | 8 |   | 4 | 5 |
| 4 |   |   |   | 5 |   |   | 2 |   |
| 6 |   |   |   | 4 |   | 5 |   |   |
|   | 5 |   | 3 |   |   |   |   |   |

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|---|---|--|---|---|---|--|---|---|
| 2 |   |  | 9 |   | 1 |  |   | 3 |
| 5 |   |  |   |   |   |  |   | 7 |
|   |   |  | 4 |   | 8 |  |   |   |
|   | 6 |  |   |   |   |  | 4 |   |
|   |   |  | 1 | 3 | 5 |  |   |   |
|   | 8 |  |   |   |   |  | 9 |   |
|   |   |  | 3 |   | 9 |  |   |   |
| 8 |   |  |   |   |   |  |   | 6 |
| 9 |   |  | 5 |   | 2 |  |   | 8 |

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 1 |   |   | 6 |  | 8 |   |   | 2 |
|   |   |   |   |  |   |   | 3 |   |
| 4 | 3 | 6 | 1 |  |   |   |   |   |
|   |   | 5 |   |  |   |   | 9 |   |
|   |   |   | 4 |  | 2 |   |   |   |
|   | 4 |   |   |  |   | 8 |   |   |
|   |   |   |   |  | 9 | 1 | 7 | 5 |
|   | 9 |   |   |  |   |   |   |   |
| 6 |   |   | 2 |  | 7 |   |   | 9 |

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|---|---|---|---|---|---|---|---|---|
|   |   | 6 | 7 |   | 3 | 5 |   |   |
| 8 |   |   |   | 1 |   |   |   | 9 |
|   | 4 |   |   |   |   |   | 8 |   |
|   |   | 2 |   |   |   | 7 |   |   |
|   |   |   | 9 | 2 | 4 |   |   |   |
|   |   | 4 |   |   |   | 9 |   |   |
|   | 2 |   |   |   |   |   | 1 |   |
| 7 |   |   |   | 4 |   |   |   | 3 |
|   |   | 9 | 1 |   | 6 | 4 |   |   |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   | 7 | 8 |   |
|   | 6 |   | 7 | 1 |   |   | 4 |   |
| 5 |   |   |   | 2 |   |   |   |   |
|   |   | 2 |   |   | 1 |   |   |   |
|   | 8 |   | 5 |   | 2 |   | 3 |   |
|   |   |   | 4 |   |   | 6 |   |   |
|   |   |   |   | 8 |   |   |   | 1 |
|   | 4 |   |   | 9 | 7 |   | 2 |   |
|   | 3 | 6 |   |   |   |   |   |   |



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|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |
|   | 3 |   | 1 |   | 4 |   | 5 |   |
|   | 4 | 7 |   | 6 |   | 8 | 3 |   |
| 6 |   |   | 8 |   | 2 |   |   | 7 |
|   |   |   |   |   |   |   |   |   |
| 4 |   |   | 7 |   | 9 |   |   | 2 |
|   | 7 | 5 |   | 3 |   | 6 | 2 |   |
|   | 8 |   | 9 |   | 7 |   | 1 |   |
|   |   |   |   |   |   |   |   |   |

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|---|---|---|---|---|---|---|---|---|
| 8 |   |   | 2 | 1 |   |   |   |   |
|   |   |   | 4 |   |   | 1 | 9 |   |
|   |   | 7 |   |   |   |   |   |   |
| 6 |   |   |   |   | 8 | 9 |   | 3 |
| 3 |   |   |   |   |   |   |   | 5 |
| 7 |   | 8 | 6 |   |   |   |   | 1 |
|   |   |   |   |   |   | 8 |   |   |
|   | 1 | 3 |   |   | 5 |   |   |   |
|   |   |   |   | 6 | 4 |   |   | 7 |

|  |   |   |   |   |   |   |   |  |
|--|---|---|---|---|---|---|---|--|
|  |   | 2 |   | 9 | 3 |   | 8 |  |
|  | 7 | 4 |   |   | 6 |   |   |  |
|  |   |   | 4 |   |   |   | 5 |  |
|  |   |   |   |   |   |   | 2 |  |
|  |   | 1 | 3 |   | 8 | 5 |   |  |
|  | 9 |   |   |   |   |   |   |  |
|  | 6 |   |   |   | 4 |   |   |  |
|  |   |   | 9 |   |   | 8 | 7 |  |
|  | 5 |   | 8 | 7 |   | 2 |   |  |

|   |  |   |   |   |   |   |  |   |
|---|--|---|---|---|---|---|--|---|
| 1 |  |   | 5 |   |   |   |  |   |
| 6 |  |   | 2 | 1 |   | 5 |  |   |
|   |  |   | 8 |   | 9 |   |  | 2 |
|   |  | 8 |   |   |   | 3 |  |   |
|   |  | 7 |   |   |   | 1 |  |   |
|   |  | 5 |   |   |   | 9 |  |   |
| 4 |  |   | 3 |   | 8 |   |  |   |
|   |  | 3 |   | 2 | 1 |   |  | 9 |
|   |  |   |   |   | 4 |   |  | 7 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 3 | 6 | 1 |   |   | 4 |   |   |   |
|   |   |   | 3 |   |   |   |   | 6 |
|   | 7 |   |   |   |   |   |   | 9 |
|   |   | 2 |   | 1 |   |   | 5 |   |
|   |   | 9 |   |   |   | 6 |   |   |
|   | 5 |   |   | 2 |   | 8 |   |   |
| 6 |   |   |   |   |   |   | 1 |   |
| 8 |   |   |   |   | 7 |   |   |   |
|   |   |   | 9 |   |   | 5 | 6 | 4 |

# **Solutions**



1

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 8 | 4 | 9 | 5 | 6 | 7 | 3 | 2 | 1 |
| 7 | 6 | 3 | 1 | 8 | 2 | 4 | 5 | 9 |
| 1 | 2 | 5 | 3 | 4 | 9 | 6 | 7 | 8 |
| 6 | 5 | 1 | 7 | 9 | 3 | 2 | 8 | 4 |
| 3 | 9 | 2 | 8 | 5 | 4 | 7 | 1 | 6 |
| 4 | 8 | 7 | 6 | 2 | 1 | 5 | 9 | 3 |
| 2 | 1 | 8 | 4 | 3 | 5 | 9 | 6 | 7 |
| 5 | 3 | 6 | 9 | 7 | 8 | 1 | 4 | 2 |
| 9 | 7 | 4 | 2 | 1 | 6 | 8 | 3 | 5 |

2

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 7 | 1 | 4 | 3 | 8 | 5 | 2 | 9 | 6 |
| 8 | 5 | 9 | 2 | 6 | 7 | 1 | 3 | 4 |
| 3 | 6 | 2 | 9 | 1 | 4 | 8 | 7 | 5 |
| 5 | 9 | 7 | 8 | 4 | 1 | 6 | 2 | 3 |
| 1 | 4 | 6 | 7 | 2 | 3 | 5 | 8 | 9 |
| 2 | 8 | 3 | 6 | 5 | 9 | 4 | 1 | 7 |
| 6 | 7 | 5 | 1 | 9 | 8 | 3 | 4 | 2 |
| 4 | 3 | 1 | 5 | 7 | 2 | 9 | 6 | 8 |
| 9 | 2 | 8 | 4 | 3 | 6 | 7 | 5 | 1 |



3

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 9 | 1 | 8 | 2 | 6 | 7 | 5 | 4 | 3 |
| 4 | 3 | 2 | 8 | 5 | 1 | 6 | 9 | 7 |
| 6 | 7 | 5 | 9 | 3 | 4 | 1 | 8 | 2 |
| 7 | 9 | 1 | 5 | 8 | 6 | 2 | 3 | 4 |
| 5 | 2 | 4 | 7 | 9 | 3 | 8 | 6 | 1 |
| 3 | 8 | 6 | 1 | 4 | 2 | 7 | 5 | 9 |
| 2 | 6 | 9 | 3 | 7 | 8 | 4 | 1 | 5 |
| 8 | 5 | 7 | 4 | 1 | 9 | 3 | 2 | 6 |
| 1 | 4 | 3 | 6 | 2 | 5 | 9 | 7 | 8 |

4

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 9 | 6 | 2 | 1 | 7 | 3 | 8 | 5 | 4 |
| 5 | 7 | 4 | 2 | 8 | 9 | 1 | 6 | 3 |
| 1 | 3 | 8 | 5 | 6 | 4 | 7 | 9 | 2 |
| 4 | 2 | 6 | 8 | 5 | 7 | 3 | 1 | 9 |
| 3 | 8 | 9 | 4 | 2 | 1 | 6 | 7 | 5 |
| 7 | 1 | 5 | 9 | 3 | 6 | 4 | 2 | 8 |
| 8 | 4 | 1 | 7 | 9 | 2 | 5 | 3 | 6 |
| 2 | 5 | 3 | 6 | 1 | 8 | 9 | 4 | 7 |
| 6 | 9 | 7 | 3 | 4 | 5 | 2 | 8 | 1 |

5

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 6 | 8 | 5 | 2 | 4 | 3 | 7 | 1 | 9 |
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| 9 | 7 | 2 | 3 | 1 | 5 | 4 | 6 | 8 |
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| 4 | 3 | 8 | 5 | 2 | 1 | 6 | 9 | 7 |
| 2 | 1 | 7 | 6 | 9 | 4 | 8 | 5 | 3 |

6

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 2 | 8 | 7 | 9 | 3 | 1 | 6 | 4 | 5 |
| 6 | 7 | 4 | 1 | 5 | 3 | 8 | 9 | 2 |
| 8 | 3 | 5 | 7 | 2 | 9 | 4 | 1 | 6 |
| 9 | 1 | 2 | 4 | 8 | 6 | 3 | 5 | 7 |

7

|   |   |   |   |   |   |   |   |   |
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| 2 | 4 | 8 | 7 | 3 | 6 | 5 | 1 | 9 |
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| 8 | 9 | 6 | 4 | 1 | 5 | 2 | 3 | 7 |
| 4 | 3 | 1 | 9 | 2 | 7 | 8 | 5 | 6 |

8

|   |   |   |   |   |   |   |   |   |
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| 2 | 7 | 5 | 4 | 8 | 1 | 9 | 6 | 3 |
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| 5 | 3 | 2 | 6 | 4 | 8 | 1 | 9 | 7 |
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9

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 2 | 7 | 9 | 5 | 4 | 3 | 1 | 6 | 8 |
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| 6 | 2 | 7 | 8 | 9 | 1 | 5 | 4 | 3 |

10

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 7 | 9 | 2 | 5 | 8 | 1 | 3 | 6 | 4 |

11

|   |   |   |   |   |   |   |   |   |
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| 6 | 7 | 5 | 4 | 1 | 2 | 3 | 9 | 8 |
| 4 | 3 | 9 | 8 | 5 | 7 | 1 | 6 | 2 |

12

|   |   |   |   |   |   |   |   |   |
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| 6 | 2 | 9 | 5 | 3 | 4 | 1 | 8 | 7 |

13

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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14

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 4 | 7 | 8 | 2 | 3 | 1 | 9 | 6 | 5 |
| 1 | 5 | 2 | 6 | 7 | 9 | 4 | 3 | 8 |

15

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 9 | 7 | 4 | 8 | 3 | 1 | 5 | 6 | 2 |
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| 8 | 3 | 9 | 4 | 1 | 7 | 6 | 2 | 5 |

16

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| 2 | 7 | 5 | 9 | 6 | 8 | 3 | 1 | 4 |

17

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 8 | 1 | 6 | 9 | 7 | 2 | 5 | 3 | 4 |
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| 6 | 4 | 2 | 5 | 9 | 1 | 7 | 8 | 3 |
| 1 | 7 | 5 | 8 | 2 | 3 | 4 | 6 | 9 |

18

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 6 | 9 | 5 | 3 | 2 | 4 | 8 | 1 | 7 |
| 7 | 4 | 6 | 9 | 3 | 2 | 1 | 8 | 5 |
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| 9 | 2 | 3 | 1 | 5 | 8 | 7 | 6 | 4 |
| 2 | 3 | 9 | 5 | 4 | 1 | 6 | 7 | 8 |
| 8 | 6 | 1 | 2 | 9 | 7 | 5 | 4 | 3 |
| 5 | 7 | 4 | 8 | 6 | 3 | 9 | 2 | 1 |



19

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 4 | 5 | 9 | 7 | 3 | 2 | 6 |
| 3 | 5 | 6 | 8 | 1 | 2 | 9 | 4 | 7 |
| 2 | 9 | 7 | 3 | 6 | 4 | 5 | 1 | 8 |
| 4 | 3 | 8 | 2 | 7 | 5 | 6 | 9 | 1 |
| 9 | 7 | 1 | 4 | 3 | 6 | 2 | 8 | 5 |
| 5 | 6 | 2 | 9 | 8 | 1 | 4 | 7 | 3 |
| 1 | 8 | 5 | 6 | 2 | 9 | 7 | 3 | 4 |
| 6 | 2 | 3 | 7 | 4 | 8 | 1 | 5 | 9 |
| 7 | 4 | 9 | 1 | 5 | 3 | 8 | 6 | 2 |

20

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 2 | 3 | 1 | 7 | 5 | 9 | 8 | 4 | 6 |
| 9 | 8 | 7 | 6 | 3 | 4 | 1 | 5 | 2 |
| 7 | 9 | 6 | 1 | 4 | 5 | 2 | 8 | 3 |
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| 3 | 1 | 5 | 4 | 7 | 6 | 9 | 2 | 8 |
| 8 | 7 | 2 | 5 | 9 | 3 | 6 | 1 | 4 |
| 6 | 4 | 9 | 2 | 1 | 8 | 5 | 3 | 7 |

21

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 2 | 9 | 3 | 4 | 5 | 6 | 7 | 8 | 1 |
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| 4 | 2 | 7 | 8 | 1 | 5 | 6 | 9 | 3 |
| 6 | 8 | 5 | 9 | 3 | 2 | 4 | 1 | 7 |
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22

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 1 | 4 | 9 | 3 | 6 | 5 | 8 | 2 | 7 |
| 5 | 6 | 8 | 7 | 4 | 2 | 1 | 3 | 9 |
| 8 | 5 | 1 | 6 | 9 | 3 | 7 | 4 | 2 |
| 2 | 9 | 4 | 8 | 1 | 7 | 3 | 6 | 5 |
| 7 | 3 | 6 | 2 | 5 | 4 | 9 | 1 | 8 |

23

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 4 | 3 | 7 | 9 | 1 | 6 | 8 | 5 | 2 |
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| 1 | 9 | 2 | 6 | 7 | 5 | 3 | 8 | 4 |
| 7 | 4 | 5 | 8 | 3 | 9 | 1 | 2 | 6 |

24

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 1 | 9 | 4 | 7 | 3 | 6 | 2 | 8 |
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| 6 | 7 | 3 | 8 | 2 | 5 | 9 | 1 | 4 |
| 3 | 9 | 2 | 5 | 6 | 1 | 4 | 8 | 7 |
| 8 | 4 | 1 | 9 | 3 | 7 | 5 | 6 | 2 |
| 7 | 5 | 6 | 2 | 8 | 4 | 1 | 9 | 3 |
| 1 | 3 | 7 | 6 | 5 | 2 | 8 | 4 | 9 |
| 2 | 8 | 4 | 7 | 1 | 9 | 3 | 5 | 6 |
| 9 | 6 | 5 | 3 | 4 | 8 | 2 | 7 | 1 |

25

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 5 | 3 | 2 | 4 | 6 | 7 | 8 | 9 | 1 |
| 7 | 1 | 8 | 2 | 9 | 5 | 3 | 6 | 4 |
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| 8 | 4 | 1 | 9 | 2 | 6 | 5 | 7 | 3 |
| 6 | 7 | 3 | 5 | 4 | 8 | 9 | 1 | 2 |
| 3 | 2 | 7 | 1 | 5 | 9 | 4 | 8 | 6 |
| 9 | 8 | 4 | 6 | 3 | 2 | 1 | 5 | 7 |
| 1 | 5 | 6 | 8 | 7 | 4 | 2 | 3 | 9 |

26

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 8 | 9 | 7 | 5 | 4 | 3 | 2 | 6 | 1 |
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| 5 | 7 | 3 | 2 | 6 | 4 | 8 | 1 | 9 |
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| 6 | 8 | 1 | 4 | 2 | 5 | 3 | 9 | 7 |
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27

|   |   |   |   |   |   |   |   |   |
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| 9 | 4 | 1 | 7 | 2 | 3 | 5 | 6 | 8 |
| 8 | 3 | 7 | 4 | 5 | 6 | 9 | 1 | 2 |

28

|   |   |   |   |   |   |   |   |   |
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| 4 | 6 | 5 | 2 | 9 | 1 | 7 | 3 | 8 |
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| 6 | 2 | 4 | 5 | 7 | 9 | 8 | 1 | 3 |
| 8 | 9 | 1 | 3 | 2 | 6 | 4 | 7 | 5 |
| 7 | 5 | 3 | 1 | 4 | 8 | 2 | 6 | 9 |

29

|   |   |   |   |   |   |   |   |   |
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| 4 | 6 | 7 | 9 | 5 | 1 | 3 | 2 | 8 |
| 2 | 7 | 6 | 1 | 9 | 4 | 8 | 5 | 3 |
| 3 | 5 | 4 | 8 | 7 | 2 | 6 | 1 | 9 |
| 8 | 1 | 9 | 5 | 6 | 3 | 2 | 4 | 7 |
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| 7 | 4 | 2 | 6 | 3 | 9 | 5 | 8 | 1 |
| 6 | 8 | 3 | 7 | 1 | 5 | 4 | 9 | 2 |

30

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 3 | 7 | 1 | 8 | 6 | 9 | 5 | 2 | 4 |
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| 4 | 9 | 5 | 1 | 3 | 2 | 7 | 8 | 6 |

31

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 1 | 9 | 7 | 6 | 3 | 4 | 8 | 2 | 5 |
| 8 | 5 | 3 | 2 | 1 | 7 | 9 | 4 | 6 |
| 3 | 7 | 2 | 5 | 4 | 9 | 6 | 1 | 8 |
| 4 | 1 | 9 | 8 | 2 | 6 | 5 | 7 | 3 |
| 6 | 8 | 5 | 3 | 7 | 1 | 4 | 9 | 2 |

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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 6 | 9 | 7 | 4 | 1 | 8 | 3 | 2 | 5 |
| 8 | 3 | 5 | 7 | 9 | 2 | 4 | 6 | 1 |
| 2 | 4 | 1 | 5 | 6 | 3 | 9 | 7 | 8 |

33

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 7 | 9 | 8 | 4 | 5 | 6 | 1 | 3 | 2 |
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34

|   |   |   |   |   |   |   |   |   |
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| 6 | 5 | 8 | 7 | 4 | 3 | 9 | 1 | 2 |
| 3 | 7 | 2 | 6 | 1 | 9 | 8 | 4 | 5 |
| 4 | 1 | 9 | 5 | 2 | 8 | 3 | 7 | 6 |



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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 5 | 3 | 7 | 2 | 9 | 4 | 1 | 8 | 6 |
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| 4 | 6 | 8 | 3 | 1 | 5 | 2 | 9 | 7 |

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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 4 | 2 | 9 | 6 | 3 | 1 | 5 | 8 | 7 |
| 7 | 1 | 3 | 8 | 4 | 5 | 6 | 2 | 9 |
| 6 | 8 | 5 | 2 | 7 | 9 | 3 | 1 | 4 |

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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 6 | 7 | 4 | 3 | 5 | 8 | 2 | 9 | 1 |
| 5 | 3 | 2 | 4 | 1 | 9 | 8 | 7 | 6 |
| 1 | 9 | 8 | 6 | 2 | 7 | 3 | 4 | 5 |

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|   |   |   |   |   |   |   |   |   |
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| 2 | 9 | 7 | 8 | 4 | 3 | 5 | 1 | 6 |
| 3 | 6 | 4 | 1 | 7 | 5 | 8 | 9 | 2 |
| 4 | 8 | 5 | 7 | 6 | 1 | 9 | 2 | 3 |
| 7 | 1 | 3 | 2 | 5 | 9 | 6 | 4 | 8 |
| 9 | 2 | 6 | 4 | 3 | 8 | 7 | 5 | 1 |

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|   |   |   |   |   |   |   |   |   |
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| 3 | 6 | 5 | 9 | 4 | 2 | 1 | 7 | 8 |
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| 7 | 5 | 3 | 6 | 1 | 4 | 8 | 2 | 9 |
| 1 | 9 | 8 | 2 | 5 | 3 | 4 | 6 | 7 |

40

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
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| 3 | 1 | 6 | 5 | 2 | 7 | 9 | 8 | 4 |
| 8 | 7 | 4 | 6 | 1 | 9 | 5 | 2 | 3 |
| 2 | 8 | 7 | 4 | 6 | 1 | 3 | 5 | 9 |
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| 7 | 4 | 8 | 9 | 5 | 6 | 2 | 3 | 1 |

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| 4 | 8 | 1 | 9 | 2 | 7 | 3 | 5 | 6 |

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| 2 | 8 | 6 | 4 | 3 | 1 | 5 | 9 | 7 |
| 5 | 7 | 1 | 9 | 6 | 8 | 2 | 4 | 3 |

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| 6 | 3 | 9 | 7 | 8 | 2 | 1 | 5 | 4 |

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| 6 | 7 | 9 | 2 | 4 | 1 | 3 | 8 | 5 |

45

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| 1 | 2 | 8 | 5 | 3 | 9 | 4 | 6 | 7 |

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| 5 | 6 | 3 | 4 | 2 | 9 | 1 | 8 | 7 |

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| 6 | 9 | 8 | 2 | 4 | 3 | 7 | 5 | 1 |

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| 5 | 8 | 4 | 1 | 2 | 9 | 3 | 6 | 7 |

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| 5 | 6 | 8 | 3 | 2 | 7 | 9 | 1 | 4 |

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| 8 | 6 | 4 | 1 | 9 | 5 | 3 | 2 | 7 |

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| 4 | 3 | 6 | 5 | 2 | 8 | 9 | 1 | 7 |
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| 5 | 1 | 8 | 9 | 3 | 7 | 6 | 4 | 2 |

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| 1 | 7 | 9 | 5 | 3 | 2 | 8 | 6 | 4 |
| 2 | 3 | 4 | 1 | 8 | 6 | 9 | 7 | 5 |

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| 2 | 5 | 9 | 4 | 3 | 6 | 7 | 8 | 1 |
| 6 | 4 | 1 | 8 | 2 | 7 | 5 | 3 | 9 |

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| 7 | 6 | 1 | 5 | 9 | 3 | 4 | 2 | 8 |
| 4 | 2 | 5 | 7 | 8 | 6 | 1 | 9 | 3 |

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| 3 | 2 | 8 | 1 | 6 | 7 | 5 | 4 | 9 |
| 4 | 1 | 7 | 8 | 5 | 9 | 3 | 6 | 2 |

59

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60

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| 5 | 4 | 7 | 9 | 1 | 6 | 8 | 3 | 2 |
| 9 | 3 | 2 | 8 | 5 | 4 | 7 | 6 | 1 |

61

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| 3 | 7 | 8 | 6 | 4 | 1 | 5 | 2 | 9 |
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65

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69

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70

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| 8 | 1 | 9 | 6 | 5 | 3 | 4 | 7 | 2 |

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| 8 | 1 | 5 | 4 | 2 | 7 | 3 | 9 | 6 |
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74

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| 7 | 1 | 8 | 9 | 6 | 2 | 4 | 5 | 3 |

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81

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